

KAAN MENEKŞEDAĞ

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Files & Presentations: kaanm.github.io | TÜBİTAK ID: TBTK-0045-2594



SUMMARY

Electrical-Electronics Engineering graduate from METU with a background in computational mathematics and software development focused on algorithms and data. At Özyeğin University, I developed GPU-accelerated numerical solvers using C++/CUDA, achieving performance gains of up to 80-90x, and published this work at international conferences. As co-founder of a TÜBİTAK 1512 BiGG-supported startup, I managed test and software engineering processes; I later worked independently in financial markets using quantitative/algorithmic methods. I bring hands-on experience with Python, C++/CUDA, and MATLAB across data analysis, automation, and algorithm development.

WORK EXPERIENCE

Independent Investor / Trader — Freelance

2021 – Present | Bursa, Turkey

- Developed and applied independent investment strategies in the BIST and derivatives markets using quantitative and algorithmic analysis methods.
- Automated data collection and analysis workflows by building Google Sheets/Apps Script-based tools (TEFAS API integration, fund price and return tracking).
- Carried out risk management and disciplined decision-making with a data-driven approach.

Co-Founder — Dipol Teknoloji ve Mühendislik Ltd. Şti.

2019 – 2021 | Istanbul, Turkey

- As co-founder of the TÜBİTAK 1512 BiGG-supported "Dipol Teknoloji" venture, led the development of automated design support software for solenoid actuator design.
- Carried out test and software engineering tasks, including setting up test rigs, running model validation experiments, and developing software/interfaces. TÜBİTAK 1512 Project No: 2180689
- Managed product development, TÜBİTAK project reporting, and technical documentation processes.

Research and Teaching Assistant — Özyeğin University

2016 – 2019 | Istanbul, Turkey

- Developed a GPU-accelerated Monte Carlo ray tracing solver based on Nvidia OptiX using C++ and CUDA, and integrated it into the open-source OpenFOAM library. TÜBİTAK 3501 Project No: 117M420
- Achieved up to 80-90x speed-up over CPU with the developed heat transfer (radiation-conduction) solver; validated results against reference solutions in the literature.
- Applied the developed solver to industrial problems with complex geometries (e.g., glass annealing furnace simulation), modeling combined convection-conduction-radiation heat transfer.
- Presented and published this work at 3 international conferences (Eurotherm Seminar, Parallel CFD, Şişecam Glass Symposium).

Project Intern — Turkish Aerospace Industries (TAI)

Jun 2015 – Aug 2015 | Ankara, Turkey

- Worked as a co-op intern in Telemetry and Database Engineering within the Flight and Flight Test Operations department.

Intern — Uludağ Elektrik Dağıtım A.Ş.

Jul 2014 – Aug 2014 | Bursa, Turkey

- Completed a technical internship in the Fault Maintenance and Repair department.

EDUCATION

Özyeğin University

2016 – 2019 | Istanbul, Turkey

M.S. in Mechanical Engineering, Thesis Track (Withdrawn – Completed Except Thesis) — GPA: 3.28/4.00

Middle East Technical University (METU)

2011 – 2016 | Ankara, Turkey

B.S. in Electrical-Electronics Engineering — GPA: 2.33/4.00

SKILLS

Programming: Python, C++, CUDA, MATLAB

Tools: Excel / Google Sheets (Automation), SketchUp, FreeCAD, OpenFOAM

Driver License: A2B

LANGUAGES

English: C1–C2 (Advanced)

German: A2 (Basic)

Spanish: A2 (Basic)

Chinese: A1 (Beginner)